

## Series. Power Series

$(X_n)$  sequence

$$\sum_{n=1}^{\infty} X_n = X_1 + X_2 + \dots + X_m \rightarrow \text{a series} \leftarrow \sum_{n \geq 1} X_n$$

$$S_m: \sum_{k=1}^m X_k \rightarrow \text{partial sum of a series}$$

When does a series converge? When the index  $n \rightarrow \infty$

$$\sum_{n=1}^{\infty} X_n \text{ converges if } S_n \text{ converges.}$$

The geometric series  $\sum_{n=0}^{\infty} 2^n$  converges if  $|2| < 1$

$$\sum_{n=0}^{\infty} 2^n = \frac{1}{1-2}$$

↳ Proof:  $S_m = 1 + 2 + 2^2 + \dots + 2^m$  partial sum.

$$= 1 \cdot \frac{2^{m+1} - 1}{2 - 1}$$

$$= \frac{1 - 2^{m+1}}{1 - 2}$$

$\rightarrow 0$  if  $|2| < 1$   
 $\rightarrow \infty$  if  $|2| \geq 1$

$$\Rightarrow \lim_n S_m = \frac{1}{1-2} \text{ if } |2| < 1. \text{ g.e.d.}$$

! If  $|2| \geq 1$ , then  $2^{m+1} \xrightarrow{(\rightarrow \infty)}$  is divergent, and the series is also divergent.  $(\rightarrow \infty)$

divergent seq. does not necessarily mean divergent series.

The harmonic series  $\sum_{n=0}^{\infty} \frac{1}{n} = \infty$  diverges since  $S_m$  is not a Cauchy seq.

If  $\sum_{n=1}^{\infty} X_n$  is convergent  $\Rightarrow \lim_n X_n = 0$ .

If  $\lim_n X_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} X_n$  divergent.

↳ Proof: Let  $S_m = X_1 + \dots + X_m \rightarrow$  partial sum

$(S_m)$  convergent (we say)

$$X_m = S_m - S_{m-1}$$

$$\lim_n X_n = \lim_n (S_m - S_{m-1}) = \lim_n S_m - \lim_n S_{m-1} = 0$$

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \text{ (converges)}$$

$\sum_{n=1}^{\infty} x_n$  with positive terms converges if it is bounded.

↳ Proof: Let  $S_n = \overbrace{x_1 + \dots + x_{n-1}}^{S_{n-1}} + \overbrace{x_n}^{>0} > S_{n-1}$   
 $\Rightarrow (S_n)$  increasing  
 $x_n > 0$

So  $(S_n)$  is convergent  $(\Rightarrow)$   $(S_n)$  bounded.

### Theorem ① (Comparison test)

$\sum_{n=1}^{\infty} x_n, \sum_{n=1}^{\infty} y_n, \quad x_n \leq y_n, \quad (\forall) n \geq n_0$

- $x_n \xrightarrow{(\rightarrow \infty)}$  diverges  $\Rightarrow y_n \xrightarrow{(\rightarrow \infty)}$  diverges
- $y_n$  converges  $\Rightarrow$  bounded  $\Rightarrow x_n$  converges  $\Rightarrow$  bounded

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \infty \text{ if } p \leq 1$$

↳ Proof:  $n^p \leq n, \quad (\forall) p \leq 1$

$$\frac{1}{n^p} \geq \frac{1}{n} \quad | \quad \sum$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \geq \sum_{n=1}^{\infty} \frac{1}{n} = \infty \text{ divergent.}$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} = \infty \text{ divergent}$$

### Theorem ② (Comparing 2 series)

$$\sum_{n=1}^{\infty} x_n, \sum_{n=1}^{\infty} y_n$$

$\lim_n \frac{x_n}{y_n} = l$

- finite  $l \in (0, \infty)$ :  $x_n = l \cdot y_n \Rightarrow x_n, y_n$  have the same nature.
- $l = 0$  Th(1)  $\Rightarrow$  if  $y_n$  converges,  $x_n$  converges.  
 $x_n < y_n$
- $l = \infty$  Th(1)  $\Rightarrow$  if  $y_n$  diverges,  $x_n$  diverges.  
 $y_n < x_n$

$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left( \frac{1}{n} - \frac{1}{n+1} \right) = 1$  telescoping series  $\rightarrow$  first one and last one remain

$\hookrightarrow S_N = \sum_{n=1}^N \frac{1}{n(n+1)} = \sum_{n=1}^N \frac{n+1-n}{n(n+1)} = \sum_{n=1}^N \left( \frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{N+1}$  the partial sum of the series

$\hookrightarrow$  The sum of the entire series:

$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \left( 1 - \frac{1}{N+1} \right) = 1 - 0 = 1. \Rightarrow \text{convergent.}$

Ex:  $\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} x_n$  convergent?

- we could make use of  $y_n$  convergent -

$\Rightarrow$  Comparison:  $\lim_n \frac{x_n}{y_n} = \lim_n \frac{\frac{1}{n^2}}{\frac{1}{n(n+1)}} = \lim_n \frac{n(n+1)}{n^2} = \lim_n \frac{n+1}{n} = 1 \in \mathbb{R}$

$1 \in (0, \infty) \Rightarrow x_n, y_n$  have the same nature.

$\Rightarrow x_n$  convergent, "like"  $y_n$ .

### Theorem (3) (Ratio test)

$\sum_{n=1}^{\infty} x_n$

$\lim_n \frac{x_{n+1}}{x_n} = l$

- $l < 1 \Rightarrow \sum_{n=1}^{\infty} x_n$  convergent
- $l > 1 \Rightarrow \sum_{n=1}^{\infty} x_n$  divergent.
- $l = 1 \Rightarrow$  inconclusive  $\rightarrow$  Cauchy ...  
 $\rightarrow$  Raabe ...

$\hookrightarrow$  Proof:  $\lim_n \frac{x_{n+1}}{x_n} = l < 1 \Rightarrow \sum x_n$  convergent

Let  $\varepsilon > 0$ .  $\exists N \in \mathbb{N}$  s.t.  $\left| \frac{x_{n+1}}{x_n} - l \right| < \varepsilon, (\forall) n \geq N$ .

$$\underbrace{1+\varepsilon}_<1 = \underbrace{q}_{>0} < 1$$

$$\Rightarrow -\varepsilon < \frac{x_{n+1}}{x_n} - l < \varepsilon$$

$$\Rightarrow \frac{x_{n+1}}{x_n} < \underbrace{l + \varepsilon}_q, (\forall) n \geq N.$$

$$\Rightarrow \frac{x_{n+1}}{x_n} < q, (\forall) n \geq N$$

$$\Rightarrow x_{n+1} < q \cdot x_n, (\forall) n \geq N \quad \text{"..."} \quad \text{"..."}$$

$$\Rightarrow \underline{x_n} < q \cdot x_{n-1} < q^2 \cdot x_{n-2} < \dots < \underline{q^{n-N} \cdot x_N} \dots, (\forall) n \geq N.$$

$$\Rightarrow x_n < q^{n-N} \cdot x_N$$

$$\Rightarrow x_n < q^n \cdot \underbrace{\frac{x_N}{q^N}}_{c = ct > 0}$$

$$\Rightarrow x_n < c \cdot q^n, (\forall) n \geq N, q < 1. \quad \left| \sum_{m=N}^{\infty} \right.$$

$$\sum_{m=N}^{\infty} x_m < c \cdot \sum_{n=N}^{\infty} q^n < \infty, \text{ since } q < 1.$$

the series is bounded by which is convergent

$$\Rightarrow \sum_{n=1}^{\infty} x_n < \infty \Leftrightarrow \sum_{n=1}^{\infty} x_n \text{ is convergent}$$

$$\text{Ex: } \sum_{n=0}^{\infty} \frac{x^n}{n!} = \sum_{n=0}^{\infty} x_n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots, \text{ for } x > 0.$$

$$\text{Ratio test: } \lim_n \frac{x_{n+1}}{x_n} = \lim_n \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} = \lim_n \frac{x}{n+1} = 0 < 1.$$

$$\Rightarrow \sum_{n=0}^{\infty} x_n \text{ is convergent, } (\forall) x > 0.$$

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} + \dots = e.}$$

! If we add all the terms in the sequence  $\Rightarrow$  series  $\Rightarrow$  we get sth. bounded, not just  $\lim \rightarrow 0$  (when  $l < 1$ )

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty, \text{ but } \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

diverges

converges  $\Rightarrow$  bounded

lb by 0  
ub by  $\approx \frac{\pi^2}{6}$

### Root test:

$$\sum_{n=1}^{\infty} x_n, \quad x_n > 0$$

$$\lim_n \sqrt[n]{x_n} = l \begin{cases} l < 1 \Rightarrow \sum x_n \text{ convergent} \\ l > 1 \Rightarrow \sum x_n \text{ divergent} \\ l = 1 \Rightarrow \text{inconclusive.} \end{cases}$$

$$\Rightarrow \sum_{n \geq 0} \left( \frac{n+1}{2n+1} \right)^n \text{ converges bc } \left( \frac{1}{2} \right)^n \rightarrow 0 < 1$$

$$\sum_{n \geq 0} \frac{x^n}{n!} \text{ converges, } (\forall) x > 0.$$

$$\sum_{n \geq 0} \frac{x^n}{n!} = e^x$$

$$\hookrightarrow \frac{x_{n+1}}{x_n} = \frac{x}{n+1} \rightarrow 0 < 1 \Rightarrow \text{converges.}$$

### Cauchy condensation test (condensing to the pw of 2) (CCT)

$$(x_n) \downarrow, \quad x_n > 0$$

$$\sum_{n=1}^{\infty} x_n, \quad \sum_{n=0}^{\infty} 2^n \cdot \underbrace{x_{2^n}}$$

$\rightarrow$  have the same nature.

only  $x_1, x_2, x_4, x_8, \dots$

$$\text{Ex: } \sum_{n=1}^{\infty} \frac{1}{n \cdot \ln n} \text{ divergent.}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \Leftrightarrow p > 1.$$

$$\hookrightarrow \text{Proof: } \sum_{n=1}^{\infty} \frac{1}{n^p}, \quad \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{(2^n)^p} \text{ have the same nature. (CCT)}$$

$$\sum_{n=1}^{\infty} \frac{2^n}{(2^n)^p} = \sum_{n=1}^{\infty} \left( \frac{2}{2^p} \right)^n = \sum_{n=1}^{\infty} \underbrace{(2^{1-p})^n}_2 = \sum_{n=1}^{\infty} 2^n \quad " \div "$$

$$\sum_{n=1}^{\infty} 2^n \text{ convergent } \Leftrightarrow |2| < 1 \Rightarrow 2^{1-p} < 1$$

$$\Leftrightarrow \frac{2}{2^p} < 1$$

$$\Leftrightarrow 2^1 < 2^p$$

$$\Leftrightarrow 1 < p$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ convergent if } p > 1 \text{ g.e.d.}$$

### Raabe - Duhamel test (if $l=1$ )

$$\sum x_n$$

$$R > 1 \Rightarrow \text{convergent}$$

$$n \geq 1 \quad \lim_n n \left( \frac{x_n}{x_{n+1}} - 1 \right) = R \quad \begin{cases} R < 1 \Rightarrow \text{divergent} \end{cases}$$

Ex:  $\sum_{n \geq 0} \frac{n!}{a(a+1)(a+2)\dots(a+n)}, a > 0.$

1) Try with the Ratio test:

$$\begin{aligned} \lim_n \frac{x_{n+1}}{x_n} &= \lim_n \frac{\cancel{(n+1)!} (n+1)}{a(a+1)\cancel{(a+m)}(a+m+1)} \cdot \frac{\cancel{a(a+1)\dots(a+m)}'}{n!} = \\ &= \lim_n \frac{n+1}{a+n+1} = 1 \Rightarrow \text{inconclusive.} \end{aligned}$$

2) Try with Raabe-Duhamel test:

$$\lim_n n \left( \frac{x_n}{x_{n+1}} - 1 \right) = \lim_n n \left( \frac{a+1+a \overset{n!}{n+1}}{n+1} - 1 \right) = \lim_n n \cdot \frac{a}{n+1} = a.$$

if  $a > 1 \Rightarrow \text{conv.}$

if  $a < 1 \Rightarrow \text{div.}$

if  $a = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{n!}{1 \cdot 2 \cdot \dots \cdot n+1} = \sum_{n=1}^{\infty} \frac{n!}{(n+1)!} = \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty \Rightarrow \text{div.}$